TAMIMI Database

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Abstract

This project presents the design and implementation of a relational database system for AlTamimi Supermarkets, a large retail organization operating multiple branches across Saudi Arabia. The objective of the system is to efficiently manage core business data related to branches, departments, employees, managers, projects, suppliers, products, customers, and sales transactions while ensuring data accuracy and integrity.

The database design process began with a detailed requirements analysis, followed by the development of an Entity Relationship Diagram (ERD) to model one-to-many and manyto-many relationships within the organization. The conceptual model was then transformed into a relational schema and normalized systematically to Third Normal Form (3NF) to eliminate redundancy, prevent update anomalies, and maintain referential integrity. Special attention was given to historical and associative tables, such as department manager history and employee-project assignments, to support real business scenarios.

The final database was implemented using Oracle SQL with clearly defined primary keys, foreign keys, and constraints. Sample data and SQL queries were executed to validate the correctness of the design and demonstrate the system's ability to retrieve meaningful operational insights. Overall, the project demonstrates the practical application of database management concepts, including ER modeling, normalization, and SQL implementation, in supporting efficient data management for a real-world retail environment.

Problem Description

Al-Tamimi Supermarkets operates multiple branches across different cities in Saudi Arabia. Each branch contains several departments such as store operations, cashiers, customer service, and marketing. The company employs a large workforce, manages suppliers, sells a wide range of products categorized by type, and handles daily customer sales transactions. Additionally, the company runs internal projects and assigns employees to these projects while tracking their duration.

Managing all this information manually or using unstructured systems can lead to data redundancy, inconsistencies, difficulty in tracking relationships, and poor decisionmaking. For example, without a centralized database, it becomes difficult to determine which employees belong to which departments, which products were sold in each transaction, or how departments and branches are performing operationally.

System Requirements

The database system must be able to:

- Store branch and department information
- Maintain employee records and departmental assignments
- Track department managers over time
- Manage suppliers and product categories
- Record products and their prices
- Process customer sales transactions and sold items
- Track employee participation in internal projects
- Maintain employee dependents
- Enforce data integrity using primary and foreign keys

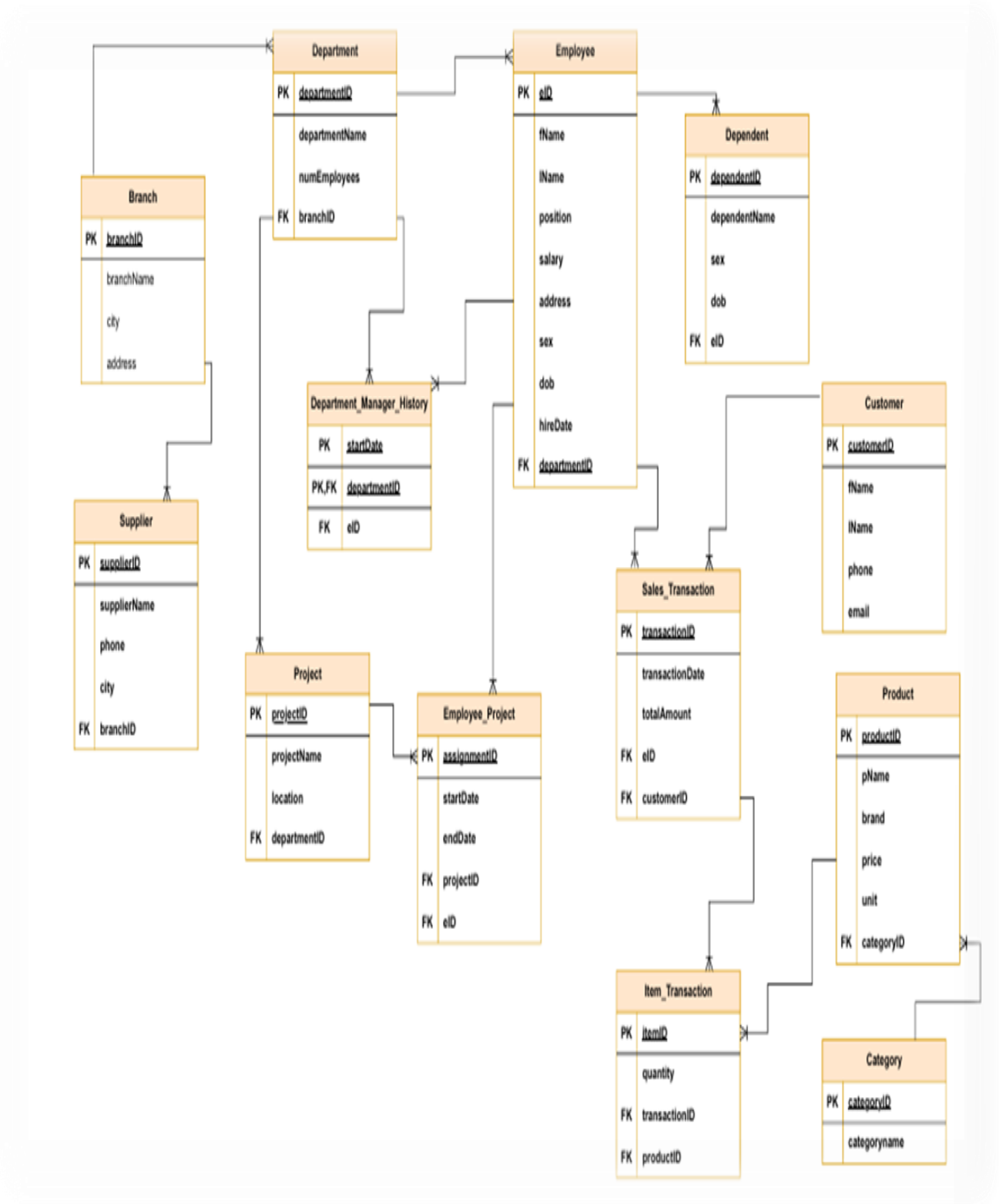
Why This Scenario Was Chosen

This scenario was chosen because it represents a realistic retail business environment that naturally includes:

- Multiple entities and relationships
- One-to-many and many-to-many relationships
- Transactional data (sales)
- Historical data (manager history, project assignments)
- Strong need for normalization and referential integrity

The Al-Tamimi case provides an ideal context to demonstrate relational database design, ERD modeling, normalization to 3NF, and Oracle SQL implementation.

Entity Relationship Diagram (ERD)



Relational Schema

Branch

- branchID (PK)
- branchName
- city
- address

Department

- departmentID (PK)
- departmentName
- numEmployees
- branchID (FK → Branch.branchID)

Employee

- eID (PK)
- fName
- lName
- position
- salary
- address
- sex
- dob
- hireDate
- departmentID (FK → Department.departmentID)

Department_Manager_History

- startDate (PK)
- departmentID (PK, FK → Department.departmentID)
- eID (FK → Employee.eID)

Project

- projectID (PK)
- projectName
- location
- departmentID (FK → Department.departmentID)

Employee_Project

- assignmentID (PK)
- startDate
- endDate
- projectID (FK → Project.projectID)
- eID (FK → Employee.eID)

Dependent

- dependentID (PK)
- dependentName
- sex
- relationship
- dob
- eID (FK → Employee.eID)

Customer

- customerID (PK)

- fName
- lName
- phone
- email

Sales_Transaction

- transactionID (PK)
- transactionDate
- totalAmount
- eID (FK → Employee.eID)
- customerID (FK → Customer.customerID)

Item_Transaction

- itemID (PK)
- quantity

transactionID (FK → Sales_Transaction.transactionID)

- productID (FK → Product.productID) **Product**
- productID (PK)
- pName
- brand
- price
- unit
- categoryID (FK → Category.categoryID) **Category**
- categoryID (PK)
- categoryName

Supplier

- supplierID (PK)
- supplierName
- phone
- city
- branchID (FK → Branch.branchID)

Normalization

The database was normalized to Third Normal Form (3NF) to eliminate redundancy and ensure data integrity. The normalization process is demonstrated below using key tables mentioned in the scenario:

Example 1: Employee Table Unnormalized

Form (UNF)

In an unnormalized design, employee records could include repeating department details such as department name and branch information, causing redundancy and transitive

dependencies; therefore, department data was separated into its own table and linked using a foreign key.

First Normal Form (1NF)

- All attributes are atomic
- No repeating groups
- Each employee has a single value per attribute

Second Normal Form (2NF)

- Employee table has a single primary key (eID)
- No partial dependency

Third Normal Form (3NF)

- No transitive dependencies
- Department data is separated into the Department table Final structure:

Employee (eID, fName, lName, position, salary, address, sex, dob, hireDate, departmentID)

Example 2: Project Table

Unnormalized Form (UNF)

In an unnormalized design project records could include department names and other department-related details stored directly within the project record, leading to data redundancy and update anomalies whenever department information changes.

First Normal Form (1NF)

- All attributes contain atomic values
- Each project is represented by a single row
- No repeating groups exist

Second Normal Form (2NF)

- The Project table has a single primary key (projectID)
- All non-key attributes depend fully on the primary key

Third Normal Form (3NF)

Department information is separated into the Department table

- The Project table references Department using a foreign key

(departmentID), eliminating transitive dependency **Final structure:**

Project (projectID, projectName, location, departmentID)

Example 3: Dependent Table

Unnormalized Form (UNF)

In an unnormalized design employee records could include multiple dependents stored in the same row (repeating groups), such as dependent1Name, dependent2Name, and so on, which leads to redundancy and makes it difficult to add or remove dependents without anomalies.

First Normal Form (1NF)

- Each dependent is stored as a separate row
- All attributes are atomic
- No repeating groups exist **Second Normal Form (2NF)**
- The Dependent table has a single primary key (dependentID)
- All non-key attributes depend fully on dependentID **Third Normal Form (3NF)**
- No transitive dependencies exist
- Dependent information is stored only in the Dependent table and linked to the employee using a foreign key (eID) **Final structure:**

Dependent (dependentID, dependentName, sex, relationship, dob, eID)

Example 4: Department Table

Unnormalized Form (UNF)

In an unnormalized design department records could include repeating branch details such as branch name, city, and address, which would lead to redundancy and update anomalies whenever branch information changes. **First Normal Form (1NF)**

- All attributes contain atomic values
- No repeating groups exist within the department records **Second Normal Form (2NF)**
- The Department table has a single primary key (departmentID)
- All non-key attributes depend fully on the primary key **Third Normal Form (3NF)**
- No transitive dependencies exist
- Branch-related information is stored in the Branch table and referenced using a foreign key **Final structure:**

Department (departmentID, departmentName, numEmployees, branchID)

Example 5: Employee_Project Table

Unnormalized Form (UNF)

In an unnormalized structure employee project assignments could include repeated employee and project details in the same table, leading to redundancy and difficulty in tracking assignment history. **First Normal Form (1NF)**

- Each assignment is stored as a separate row
- Start and end dates are stored as atomic attributes

Second Normal Form (2NF)

- The table uses a single primary key (assignmentID)

All non-key attributes depend fully on the primary key **Third Normal Form (3NF)**

- Employee and project details are stored in their respective tables
- The Employee_Project table serves as an associative table that resolves the manyto-many relationship between employees and projects **Final structure:**

Employee_Project (assignmentID, startDate, endDate, projectID, eID)

Example 6: Department_Manager_History Table

Unnormalized Form (UNF)

Originally, manager information could be stored directly inside the Department table (e.g., managerName and managerStartDate), which would overwrite old data whenever a manager changes and prevents keeping a history of previous managers. **First Normal Form (1NF)**

- Each manager assignment is stored as a separate row
- Dates and IDs are stored as atomic attributes
- No repeating groups exist **Second Normal Form (2NF)**
- The table uses a composite primary key (departmentID, startDate)
- Any non-key attribute (eID) depends on the whole composite key, not part of it **Third Normal Form (3NF)**
- No transitive dependencies exist
- Employee details remain in the Employee table, and the history table stores only the manager assignment reference (eID) with its effective start date
- **Final structure:**

Department_Manager_History (startDate, departmentID, eID)

Conclusion on Normalization

All tables satisfy the normalization requirements:

- **1NF**: All attributes are atomic and contain single values
- **2NF**: All non-key attributes are fully functionally dependent on the primary key
- **3NF**: No transitive dependencies exist among non-key attributes

Therefore, the final database design is fully normalized to Third Normal Form (3NF).

Tables in 3NF

Branch Table (3NF)

Attribute Name	Data Type	Key / Constraints
branchID	NUMBER(3)	PK
branchName	VARCHAR2(50)	NOT NULL
city	VARCHAR2(30)	NOT NULL
address	VARCHAR2(100)	-

Category Table (3NF)

Attribute Name	Data Type	Key / Constraints
categoryID	NUMBER(3)	PK
categoryName	VARCHAR2(50)	NOT NULL, UNIQUE

Department Table (3NF)

Attribute Name	Data Type	Key / Constraints
departmentID	NUMBER(3)	PK
departmentName	VARCHAR2(50)	NOT NULL
numEmployees	NUMBER(3)	DEFAULT 0, CHECK ≥ 0
branchID	NUMBER(3)	FK → Branch(branchID), NOT NULL

-	-	**UNIQUE (departmentName, branchID)**

Employee Table (3NF)

Attribute Name	Data Type	Key / Constraints
eID	NUMBER(4)	PK
fName	VARCHAR2(30)	NOT NULL
lName	VARCHAR2(30)	NOT NULL
position	VARCHAR2(50)	NOT NULL
salary	NUMBER(8,2)	NOT NULL, CHECK ≥ 0
address	VARCHAR2(100)	-
sex	CHAR(1)	NOT NULL, CHECK (M/F)
dob	DATE	NOT NULL
hireDate	DATE	NOT NULL
departmentID	NUMBER(3)	FK → Department(departmentID), NOT NULL

Project Table (3NF)

Attribute Name	Data Type	Key / Constraints
projectID	NUMBER(4)	PK
projectName	VARCHAR2(80)	NOT NULL
location	VARCHAR2(100)	-
departmentID	NUMBER(3)	FK → Department(departmentID), NOT NULL

Employee_Project Table (3NF) (Associative (Junction) Table, Many to Many)

Attribute Name	Data Type	Key / Constraints
assignmentID	NUMBER(5)	PK
startDate	DATE	NOT NULL
endDate	DATE	CHECK (endDate IS NULL OR endDate ≥ startDate)
projectID	NUMBER(4)	FK → Project(projectID), NOT NULL
eID	NUMBER(4)	FK → Employee(eID), NOT NULL
-	-	UNIQUE (eID, projectID, startDate)

Dependent Table (3NF)

Attribute Name	Data Type	Key / Constraints
dependentID	NUMBER(4)	PK
dependentName	VARCHAR2(50)	NOT NULL
sex	CHAR(1)	NOT NULL, CHECK (M/F)
relationship	VARCHAR2(20)	NOT NULL
dob	DATE	NOT NULL

eID	**NUMBER(4)**	**FK → Employee(eID), NOT NULL**

Department_Manager_History Table (3NF) (Composite Key)

Attribute Name	Data Type	Key / Constraints
startDate	**DATE**	**PK, NOT NULL**
departmentID	**NUMBER(3)**	**PK, FK → Department(departmentID), NOT NULL**
eID	**NUMBER(4)**	**FK → Employee(eID)**

Customer Table (3NF)

Attribute Name	Data Type	Key / Constraints
customerID	**NUMBER(4)**	**PK**
fName	**VARCHAR2(30)**	**NOT NULL**
lName	**VARCHAR2(30)**	**NOT NULL**
phone	**VARCHAR2(20)**	**-**
email	**VARCHAR2(100)**	**UNIQUE**

Supplier Table (3NF)

Attribute Name	Data Type	Key / Constraints
supplierID	**NUMBER(4)**	**PK**
supplierName	**VARCHAR2(80)**	**NOT NULL**
phone	**VARCHAR2(20)**	**-**
city	**VARCHAR2(30)**	**-**
branchID	**NUMBER(3)**	**FK → Branch(branchID), NOT NULL**

Product Table (3NF)

Attribute Name	Data Type	Key / Constraints
productID	NUMBER(4)	PK
pName	VARCHAR2(50)	NOT NULL
brand	VARCHAR2(50)	-
price	NUMBER(8,2)	CHECK ≥ 0
unit	VARCHAR2(20)	NOT NULL
categoryID	NUMBER(3)	FK → Category(categoryID), NOT NULL

Sales_Transaction Table (3NF)

Attribute Name	Data Type	Key / Constraints
transactionID	NUMBER(6)	PK
transactionDate	DATE	NOT NULL
totalAmount	NUMBER(10,2)	CHECK ≥ 0, NOT NULL
eID	NUMBER(4)	FK → Employee(eID), NOT NULL
customerID	NUMBER(4)	FK → Customer(customerID), NOT NULL

Item_Transaction Table (3NF)

Attribute Name	Data Type	Key / Constraints

itemID	**NUMBER(5)**	**PK**
quantity	**NUMBER(5)**	**CHECK > 0, NOT NULL**
transactionID	**NUMBER(6)**	**FK → Sales_Transaction(transactionID), NOT NULL**
productID	**NUMBER(4)**	**FK → Product(productID), NOT NULL**
-	-	**UNIQUE (transactionID, productID)**

SQL: 1.1 Codes

```
CREATE TABLE Branch (
  branchID    NUMBER(3)     CONSTRAINT pk_branch PRIMARY KEY,
  branchName  VARCHAR2(50)  CONSTRAINT nn_branch_name NOT NULL,
  city        VARCHAR2(30)  CONSTRAINT nn_branch_city NOT NULL,
  address     VARCHAR2(100));

CREATE TABLE Category (
  categoryID    NUMBER(3)     CONSTRAINT pk_category PRIMARY KEY,
  categoryName  VARCHAR2(50)  CONSTRAINT nn_category_name NOT NULL,
  CONSTRAINT uq_category_name UNIQUE (categoryName));

CREATE TABLE Department (
  departmentID    NUMBER(3)     CONSTRAINT pk_department PRIMARY KEY,
  departmentName  VARCHAR2(50)  CONSTRAINT nn_dept_name NOT NULL,
  numEmployees    NUMBER(3)     DEFAULT 0 CONSTRAINT ck_dept_numemp CHECK (numEmployees >= 0),
  branchID        NUMBER(3)     CONSTRAINT nn_dept_branch NOT NULL,
  CONSTRAINT fk_dept_branch FOREIGN KEY (branchID) REFERENCES Branch(branchID),
  CONSTRAINT uq_dept_name_branch UNIQUE (departmentName, branchID));

CREATE TABLE Employee (
  eID           NUMBER(4)     CONSTRAINT pk_employee PRIMARY KEY,
  fName         VARCHAR2(30)  CONSTRAINT nn_emp_fname NOT NULL,
  lName         VARCHAR2(30)  CONSTRAINT nn_emp_lname NOT NULL,
  position      VARCHAR2(50)  CONSTRAINT nn_emp_position NOT NULL,
  salary        NUMBER(8,2)   CONSTRAINT nn_emp_salary NOT NULL,
  address       VARCHAR2(100),
  sex           CHAR(1)       CONSTRAINT nn_emp_sex NOT NULL,
  DoB           DATE          CONSTRAINT nn_emp_dob NOT NULL,
  hireDate      DATE          CONSTRAINT nn_emp_hire NOT NULL,
  departmentID  NUMBER(3)     CONSTRAINT nn_emp_dept NOT NULL,
  CONSTRAINT fk_emp_dept FOREIGN KEY (departmentID) REFERENCES Department(departmentID),
  CONSTRAINT ck_emp_sex CHECK (sex IN ('M','F')),
  CONSTRAINT ck_emp_salary CHECK (salary >= 0));
```

Figure 1 Tables (Branch, Category, Department, Employee) CREATE codes

```
CREATE TABLE Customer (
  customerID  NUMBER(4)     CONSTRAINT pk_customer PRIMARY KEY,
  fName       VARCHAR2(30)  CONSTRAINT nn_cust_fname NOT NULL,
  lName       VARCHAR2(30)  CONSTRAINT nn_cust_lname NOT NULL,
  phone       VARCHAR2(20),
  email       VARCHAR2(100),
  CONSTRAINT uq_customer_email UNIQUE (email));

CREATE TABLE Supplier (
  supplierID    NUMBER(4)     CONSTRAINT pk_supplier PRIMARY KEY,
  supplierName  VARCHAR2(80)  CONSTRAINT nn_supplier_name NOT NULL,
  phone         VARCHAR2(20),
  city          VARCHAR2(30),
  branchID      NUMBER(3)     CONSTRAINT nn_supplier_branch NOT NULL,
  CONSTRAINT fk_supplier_branch FOREIGN KEY (branchID) REFERENCES Branch(branchID));

CREATE TABLE Product (
  productID   NUMBER(4)     CONSTRAINT pk_product PRIMARY KEY,
  pName       VARCHAR2(50)  CONSTRAINT nn_product_name NOT NULL,
  brand       VARCHAR2(50),
  price       NUMBER(8,2)   CONSTRAINT nn_product_price NOT NULL,
  unit        VARCHAR2(20)  CONSTRAINT nn_product_unit NOT NULL,
  categoryID  NUMBER(3)     CONSTRAINT nn_product_cat NOT NULL,
  CONSTRAINT fk_product_cat FOREIGN KEY (categoryID) REFERENCES Category(categoryID),
  CONSTRAINT ck_product_price CHECK (price >= 0));

CREATE TABLE Project (
  projectID     NUMBER(4)     CONSTRAINT pk_project PRIMARY KEY,
  projectName   VARCHAR2(80)  CONSTRAINT nn_project_name NOT NULL,
  location      VARCHAR2(100),
  departmentID  NUMBER(3)     CONSTRAINT nn_project_dept NOT NULL,
  CONSTRAINT fk_project_dept FOREIGN KEY (departmentID) REFERENCES Department(departmentID));
```

Figure 2 Tables (Customer, Supplier, Product, Project) CREATE Codes

```
CREATE TABLE Sales_Transaction (
  transactionID    NUMBER(6)    CONSTRAINT pk_sales_trans PRIMARY KEY,
  transactionDate  DATE         CONSTRAINT nn_trans_date NOT NULL,
  totalAmount      NUMBER(10,2) CONSTRAINT nn_trans_total NOT NULL,
  eID              NUMBER(4)    CONSTRAINT nn_trans_emp NOT NULL,
  customerID       NUMBER(4)    CONSTRAINT nn_trans_cust NOT NULL,
  CONSTRAINT fk_sale_emp  FOREIGN KEY (eID)        REFERENCES Employee(eID),
  CONSTRAINT fk_sale_cust FOREIGN KEY (customerID) REFERENCES Customer(customerID),
  CONSTRAINT ck_trans_total CHECK (totalAmount >= 0));

CREATE TABLE Department_Manager_History (
  startDate     DATE       CONSTRAINT nn_dmh_start NOT NULL,
  departmentID  NUMBER(3)  CONSTRAINT nn_dmh_dept  NOT NULL,
  eID           NUMBER(4)  CONSTRAINT nn_dmh_emp   NOT NULL,
  CONSTRAINT pk_dept_mgr PRIMARY KEY (startDate, departmentID),
  CONSTRAINT fk_dmh_dept FOREIGN KEY (departmentID) REFERENCES Department(departmentID),
  CONSTRAINT fk_dmh_emp  FOREIGN KEY (eID)         REFERENCES Employee(eID));

CREATE TABLE Employee_Project (
  assignmentID  NUMBER(5)  CONSTRAINT pk_empproj PRIMARY KEY,
  startDate     DATE       CONSTRAINT nn_empproj_start NOT NULL,
  endDate       DATE,
  projectID     NUMBER(4)  CONSTRAINT nn_empproj_proj NOT NULL,
  eID           NUMBER(4)  CONSTRAINT nn_empproj_emp  NOT NULL,
  CONSTRAINT fk_empproj_proj FOREIGN KEY (projectID) REFERENCES Project(projectID),
  CONSTRAINT fk_empproj_emp  FOREIGN KEY (eID)       REFERENCES Employee(eID),
  CONSTRAINT ck_empproj_dates CHECK (endDate IS NULL OR endDate >= startDate),
  CONSTRAINT uq_empproj UNIQUE (eID, projectID, startDate));
```

Figure 3 Tables (Sales_Tranaction, Department_Manager_History, Employee_Project) CREATE Codes

```sql
CREATE TABLE Dependent (
  dependentID    NUMBER(4)    CONSTRAINT pk_dependent PRIMARY KEY,
  dependentName  VARCHAR2(50) CONSTRAINT nn_dep_name NOT NULL,
  sex            CHAR(1)      CONSTRAINT nn_dep_sex NOT NULL,
  relationship   VARCHAR2(20) CONSTRAINT nn_dep_rel NOT NULL,
  DoB            DATE         CONSTRAINT nn_dep_dob NOT NULL,
  eID            NUMBER(4)    CONSTRAINT nn_dep_emp NOT NULL,
  CONSTRAINT fk_dep_emp FOREIGN KEY (eID) REFERENCES Employee(eID),
  CONSTRAINT ck_dep_sex CHECK (sex IN ('M','F')));

CREATE TABLE Item_Transaction (
  itemID         NUMBER(5)  CONSTRAINT pk_item_trans PRIMARY KEY,
  quantity       NUMBER(5)  CONSTRAINT nn_item_qty NOT NULL,
  transactionID  NUMBER(6)  CONSTRAINT nn_item_transid NOT NULL,
  productID      NUMBER(4)  CONSTRAINT nn_item_prodid NOT NULL,
  CONSTRAINT fk_item_trans FOREIGN KEY (transactionID) REFERENCES Sales_Transaction(transactionID),
  CONSTRAINT fk_item_prod  FOREIGN KEY (productID)     REFERENCES Product(productID),
  CONSTRAINT ck_item_qty CHECK (quantity > 0),
  CONSTRAINT uq_item_line UNIQUE (transactionID, productID));
```

Figure 4 Tables (Dependent, Item_Transaction) CREATE Codes

```sql
INSERT INTO Branch (branchID, branchName, city, address) VALUES (101, 'Granada Branch',  'Riyadh', 'Granada Mall, Gate 3');
INSERT INTO Branch (branchID, branchName, city, address) VALUES (102, 'Corniche Branch', 'Dammam', 'Corniche Road');
INSERT INTO Branch (branchID, branchName, city, address) VALUES (103, 'Nakheel Branch',  'Riyadh', 'Nakheel Mall, Gate 2');

INSERT INTO Department (departmentID, departmentName, numEmployees, branchID) VALUES (1, 'Store Operations',      12, 101);
INSERT INTO Department (departmentID, departmentName, numEmployees, branchID) VALUES (2, 'Cashiers',              18, 101);
INSERT INTO Department (departmentID, departmentName, numEmployees, branchID) VALUES (3, 'Customer Service',      6, 101);
INSERT INTO Department (departmentID, departmentName, numEmployees, branchID) VALUES (4, 'Store Operations',      10, 102);
INSERT INTO Department (departmentID, departmentName, numEmployees, branchID) VALUES (5, 'Cashiers',              15, 102);
INSERT INTO Department (departmentID, departmentName, numEmployees, branchID) VALUES (6, 'Customer Service',      5, 102);
INSERT INTO Department (departmentID, departmentName, numEmployees, branchID) VALUES (7, 'Store Operations',      11, 103);
INSERT INTO Department (departmentID, departmentName, numEmployees, branchID) VALUES (8, 'Cashiers',              16, 103);
INSERT INTO Department (departmentID, departmentName, numEmployees, branchID) VALUES (9, 'Marketing & Promotions', 6, 103);
```

Figure 5 Tables (Branch, Department) INSERT Codes

```sql
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (201, 'Ahmed', 'Salem', 'Store Operations Manager', 7500, 'Riyadh – Granada', 'M', TO_DATE('1986-04-12','YYYY-MM-DD'), TO_DATE('2019-03-01','YYYY-MM-DD'), 1);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (202, 'Rana', 'Yousif', 'Stock Clerk', 4800, 'Riyadh – Granada', 'F', TO_DATE('1995-09-20','YYYY-MM-DD'), TO_DATE('2021-06-10','YYYY-MM-DD'), 1);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (203, 'Majed', 'Saleh', 'Floor Staff', 4600, 'Riyadh – Granada', 'M', TO_DATE('1997-11-05','YYYY-MM-DD'), TO_DATE('2022-08-15','YYYY-MM-DD'), 1);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (204, 'Layla', 'Faris', 'Cashier Supervisor', 6200, 'Riyadh – Granada', 'F', TO_DATE('1992-05-10','YYYY-MM-DD'), TO_DATE('2018-09-01','YYYY-MM-DD'), 2);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (205, 'Saif', 'Hamad', 'Cashier', 4500, 'Riyadh – Granada', 'M', TO_DATE('1998-01-22','YYYY-MM-DD'), TO_DATE('2021-02-14','YYYY-MM-DD'), 2);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (206, 'Reem', 'Nasser', 'Cashier', 4450, 'Riyadh – Granada', 'F', TO_DATE('1999-07-30','YYYY-MM-DD'), TO_DATE('2022-04-01','YYYY-MM-DD'), 2);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (207, 'Nora', 'Ali', 'CS Supervisor', 6000, 'Riyadh – Granada', 'F', TO_DATE('1993-09-02','YYYY-MM-DD'), TO_DATE('2019-01-15','YYYY-MM-DD'), 3);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (208, 'Huda', 'Ibrahim', 'CS Agent', 4700, 'Riyadh – Granada', 'F', TO_DATE('1997-03-18','YYYY-MM-DD'), TO_DATE('2021-05-20','YYYY-MM-DD'), 3);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (209, 'Sami', 'Kareem', 'CS Agent', 4650, 'Riyadh – Granada', 'M', TO_DATE('1996-12-01','YYYY-MM-DD'), TO_DATE('2022-02-10','YYYY-MM-DD'), 3);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (210, 'Faisal', 'Omar', 'Store Manager', 7800, 'Dammam – Corniche', 'M', TO_DATE('1987-12-28','YYYY-MM-DD'), TO_DATE('2018-11-01','YYYY-MM-DD'), 4);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (211, 'Ali', 'Zaid', 'Receiver', 5000, 'Dammam – Corniche', 'M', TO_DATE('1994-06-09','YYYY-MM-DD'), TO_DATE('2020-07-05','YYYY-MM-DD'), 4);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (212, 'Maha', 'Adel', 'Floor Staff', 4700, 'Dammam – Corniche', 'F', TO_DATE('1998-02-09','YYYY-MM-DD'), TO_DATE('2022-03-12','YYYY-MM-DD'), 4);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (213, 'Reema', 'Majed', 'Cashier Supervisor', 6100, 'Khobar – Thuqbah', 'F', TO_DATE('1991-03-19','YYYY-MM-DD'), TO_DATE('2019-10-01','YYYY-MM-DD'), 5);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (214, 'Tariq', 'Zayed', 'Cashier', 4550, 'Dammam – Corniche', 'M', TO_DATE('1997-08-22','YYYY-MM-DD'), TO_DATE('2021-03-25','YYYY-MM-DD'), 5);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (215, 'Lina', 'Saad', 'Cashier', 4400, 'Dammam – Corniche', 'F', TO_DATE('1999-11-11','YYYY-MM-DD'), TO_DATE('2022-06-18','YYYY-MM-DD'), 5);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (216, 'Abeer', 'Hassan', 'CS Supervisor', 5900, 'Dammam – Corniche', 'F', TO_DATE('1990-10-05','YYYY-MM-DD'), TO_DATE('2018-08-10','YYYY-MM-DD'), 6);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (217, 'Omar', 'Rashid', 'CS Agent', 4700, 'Dammam – Corniche', 'M', TO_DATE('1996-05-14','YYYY-MM-DD'), TO_DATE('2020-09-30','YYYY-MM-DD'), 6);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (218, 'Dana', 'Khalil', 'CS Agent', 4650, 'Dammam – Corniche', 'F', TO_DATE('1997-09-27','YYYY-MM-DD'), TO_DATE('2021-11-22','YYYY-MM-DD'), 6);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (219, 'Sami', 'Nasser', 'Operations Supervisor', 7000, 'Riyadh – Nakheel', 'M', TO_DATE('1989-11-05','YYYY-MM-DD'), TO_DATE('2020-01-10','YYYY-MM-DD'), 7);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (220, 'Hind', 'Mousa', 'Floor Staff', 4800, 'Riyadh – Nakheel', 'F', TO_DATE('1995-04-07','YYYY-MM-DD'), TO_DATE('2021-08-15','YYYY-MM-DD'), 7);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (221, 'Yara', 'Samir', 'Inventory Clerk', 4900, 'Riyadh – Nakheel', 'F', TO_DATE('1996-01-19','YYYY-MM-DD'), TO_DATE('2022-05-03','YYYY-MM-DD'), 7);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (222, 'Nabil', 'Saleh', 'Cashier Supervisor', 6150, 'Riyadh – Nakheel', 'M', TO_DATE('1991-02-16','YYYY-MM-DD'), TO_DATE('2019-06-20','YYYY-MM-DD'), 8);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (223, 'Waleed', 'Anwar', 'Cashier', 4550, 'Riyadh – Nakheel', 'M', TO_DATE('1998-09-09','YYYY-MM-DD'), TO_DATE('2021-01-27','YYYY-MM-DD'), 8);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (224, 'Sara', 'Jalal', 'Cashier', 4450, 'Riyadh – Nakheel', 'F', TO_DATE('1999-12-24','YYYY-MM-DD'), TO_DATE('2022-07-11','YYYY-MM-DD'), 8);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (225, 'Aisha', 'Rahman', 'Marketing Coordinator', 6400, 'Riyadh – Nakheel', 'F', TO_DATE('1988-09-12','YYYY-MM-DD'), TO_DATE('2019-02-20','YYYY-MM-DD'), 9);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (226, 'Hani', 'Yusuf', 'Graphic Designer', 5500, 'Riyadh – Nakheel', 'M', TO_DATE('1994-03-03','YYYY-MM-DD'), TO_DATE('2021-04-18','YYYY-MM-DD'), 9);
INSERT INTO Employee (eID, fName, lName, position, salary, address, sex, DoB, hireDate, departmentID) VALUES (227, 'Lujain', 'Hakim', 'Social Media Specialist', 5300, 'Riyadh – Nakheel', 'F', TO_DATE('1995-08-29','YYYY-MM-DD'), TO_DATE('2022-01-05','YYYY-MM-DD'), 9);
```

Figure 6 Table (Employee) INSERT Codes

```sql
INSERT INTO Customer (customerID, fName, lName, phone, email) VALUES (801, 'Khalid', 'Omar',   '0551122334', 'khalid.o@example.com');
INSERT INTO Customer (customerID, fName, lName, phone, email) VALUES (802, 'Mariam','Hassan', '0569988776', 'mariam.h@example.com');
INSERT INTO Customer (customerID, fName, lName, phone, email) VALUES (803, 'Saad',  'Fahad',  '0547766554', 'saad.f@example.com');

INSERT INTO Supplier (supplierID, supplierName, phone, city, branchID) VALUES (501, 'FreshFarms Co.',      '0501234567', 'Riyadh', 101);
INSERT INTO Supplier (supplierID, supplierName, phone, city, branchID) VALUES (502, 'Gulf Packaging Ltd.', '0543342211', 'Dammam', 102);
INSERT INTO Supplier (supplierID, supplierName, phone, city, branchID) VALUES (503, 'CleanHome Products',  '0558899772', 'Riyadh', 103);

INSERT INTO Category (categoryID, categoryName) VALUES (1, 'Produce');
INSERT INTO Category (categoryID, categoryName) VALUES (2, 'Bakery');
INSERT INTO Category (categoryID, categoryName) VALUES (3, 'Dairy & Beverages');
INSERT INTO Category (categoryID, categoryName) VALUES (4, 'Household & Cleaning');

INSERT INTO Product (productID, pName, brand, price, unit, categoryID) VALUES (701, 'Tomatoes',         'FreshFarms', 4.50,  'Kg',     1);
INSERT INTO Product (productID, pName, brand, price, unit, categoryID) VALUES (702, 'White Bread Loaf', 'SunBake',    7.50,  'Piece',  2);
INSERT INTO Product (productID, pName, brand, price, unit, categoryID) VALUES (703, 'Milk 2L',          'DairyPure',  8.75,  'Liter',  3);
INSERT INTO Product (productID, pName, brand, price, unit, categoryID) VALUES (704, 'Floor Cleaner',    'CleanHome',  12.00, 'Bottle', 4);

INSERT INTO Project (projectID, projectName, location, departmentID) VALUES (301, 'Aisle Reorganization',           'Granada Branch – Operations', 1);
INSERT INTO Project (projectID, projectName, location, departmentID) VALUES (302, 'Cashier Queue Optimization',     'Granada Branch – Checkout',   2);
INSERT INTO Project (projectID, projectName, location, departmentID) VALUES (303, 'Customer Feedback Kiosk Setup',  'Granada Branch – Entry Area', 3);
INSERT INTO Project (projectID, projectName, location, departmentID) VALUES (304, 'Inventory Layout Improvement',   'Corniche Branch – Warehouse', 4);
INSERT INTO Project (projectID, projectName, location, departmentID) VALUES (305, 'Express Cashier Lane Setup',     'Corniche Branch – Front End', 5);
INSERT INTO Project (projectID, projectName, location, departmentID) VALUES (306, 'Customer Inquiry System Upgrade', 'Corniche Branch – CS Desk',   6);
INSERT INTO Project (projectID, projectName, location, departmentID) VALUES (307, 'Storage Area Redesign',          'Nakheel Branch – Backroom',   7);
INSERT INTO Project (projectID, projectName, location, departmentID) VALUES (308, 'Cashier Scheduling Automation',  'Nakheel Branch – Cash Office',8);
INSERT INTO Project (projectID, projectName, location, departmentID) VALUES (309, 'Weekend Deals Ad Campaign',      'Riyadh Region',               9);
```

Figure 7 Tables (Customer, Supplier, Category, Product, Project) INSERT Codes

```sql
INSERT INTO Department_Manager_History (startDate, departmentID, eID) VALUES (TO_DATE('2019-03-01','YYYY-MM-DD'), 1, 201);
INSERT INTO Department_Manager_History (startDate, departmentID, eID) VALUES (TO_DATE('2018-09-01','YYYY-MM-DD'), 2, 204);
INSERT INTO Department_Manager_History (startDate, departmentID, eID) VALUES (TO_DATE('2019-01-15','YYYY-MM-DD'), 3, 207);
INSERT INTO Department_Manager_History (startDate, departmentID, eID) VALUES (TO_DATE('2018-11-01','YYYY-MM-DD'), 4, 210);
INSERT INTO Department_Manager_History (startDate, departmentID, eID) VALUES (TO_DATE('2019-10-01','YYYY-MM-DD'), 5, 213);
INSERT INTO Department_Manager_History (startDate, departmentID, eID) VALUES (TO_DATE('2018-08-10','YYYY-MM-DD'), 6, 216);
INSERT INTO Department_Manager_History (startDate, departmentID, eID) VALUES (TO_DATE('2020-01-10','YYYY-MM-DD'), 7, 219);
INSERT INTO Department_Manager_History (startDate, departmentID, eID) VALUES (TO_DATE('2019-06-20','YYYY-MM-DD'), 8, 222);
INSERT INTO Department_Manager_History (startDate, departmentID, eID) VALUES (TO_DATE('2019-02-20','YYYY-MM-DD'), 9, 225);

INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (901, TO_DATE('2024-01-05','YYYY-MM-DD'), TO_DATE('2024-02-10','YYYY-MM-DD'), 301, 201);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (902, TO_DATE('2024-01-05','YYYY-MM-DD'), TO_DATE('2024-02-10','YYYY-MM-DD'), 301, 202);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (903, TO_DATE('2024-01-05','YYYY-MM-DD'), TO_DATE('2024-02-10','YYYY-MM-DD'), 301, 203);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (904, TO_DATE('2024-02-12','YYYY-MM-DD'), TO_DATE('2024-03-05','YYYY-MM-DD'), 302, 204);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (905, TO_DATE('2024-02-12','YYYY-MM-DD'), TO_DATE('2024-03-05','YYYY-MM-DD'), 302, 205);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (906, TO_DATE('2024-03-08','YYYY-MM-DD'), TO_DATE('2024-03-30','YYYY-MM-DD'), 303, 207);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (907, TO_DATE('2024-03-08','YYYY-MM-DD'), TO_DATE('2024-03-30','YYYY-MM-DD'), 303, 208);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (908, TO_DATE('2024-01-15','YYYY-MM-DD'), TO_DATE('2024-02-20','YYYY-MM-DD'), 304, 210);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (909, TO_DATE('2024-02-22','YYYY-MM-DD'), TO_DATE('2024-03-18','YYYY-MM-DD'), 305, 213);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (910, TO_DATE('2024-03-10','YYYY-MM-DD'), TO_DATE('2024-04-02','YYYY-MM-DD'), 306, 216);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (911, TO_DATE('2024-01-20','YYYY-MM-DD'), TO_DATE('2024-02-25','YYYY-MM-DD'), 307, 219);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (912, TO_DATE('2024-02-27','YYYY-MM-DD'), TO_DATE('2024-03-22','YYYY-MM-DD'), 308, 222);
INSERT INTO Employee_Project (assignmentID, startDate, endDate, projectID, eID) VALUES (913, TO_DATE('2024-02-01','YYYY-MM-DD'), TO_DATE('2024-02-28','YYYY-MM-DD'), 309, 225);

INSERT INTO Sales_Transaction (transactionID, transactionDate, totalAmount, eID, customerID) VALUES (1001, TO_DATE('2024-02-15','YYYY-MM-DD'), 64.50, 205, 801);
INSERT INTO Sales_Transaction (transactionID, transactionDate, totalAmount, eID, customerID) VALUES (1002, TO_DATE('2024-02-17','YYYY-MM-DD'), 22.00, 214, 802);
INSERT INTO Sales_Transaction (transactionID, transactionDate, totalAmount, eID, customerID) VALUES (1003, TO_DATE('2024-02-17','YYYY-MM-DD'), 45.75, 223, 803);
```

Figure 8 Tables (Department_Manager_History, Employee_Project, Sales_Transaction) INSERT Codes

```sql
INSERT INTO Item_Transaction (itemID, quantity, transactionID, productID) VALUES (1, 2, 1001, 702);
INSERT INTO Item_Transaction (itemID, quantity, transactionID, productID) VALUES (2, 1, 1001, 703);
INSERT INTO Item_Transaction (itemID, quantity, transactionID, productID) VALUES (3, 3, 1002, 704);
INSERT INTO Item_Transaction (itemID, quantity, transactionID, productID) VALUES (4, 1, 1003, 701);
INSERT INTO Item_Transaction (itemID, quantity, transactionID, productID) VALUES (5, 2, 1003, 702);

INSERT INTO Dependent (dependentID, dependentName, sex, relationship, DoB, eID) VALUES (1, 'Sara Ahmed',      'F', 'Daughter', TO_DATE('2020-08-14','YYYY-MM-DD'), 201);
INSERT INTO Dependent (dependentID, dependentName, sex, relationship, DoB, eID) VALUES (2, 'Mohammed Faisal', 'M', 'Son',      TO_DATE('2015-03-01','YYYY-MM-DD'), 210);
INSERT INTO Dependent (dependentID, dependentName, sex, relationship, DoB, eID) VALUES (3, 'Layan Aisha',     'F', 'Daughter', TO_DATE('2017-11-22','YYYY-MM-DD'), 225);

COMMIT;

SELECT * FROM Branch;
SELECT * FROM Department;
SELECT * FROM Employee;
SELECT * FROM Customer;
SELECT * FROM Category;
SELECT * FROM Product;
SELECT * FROM Supplier;
SELECT * FROM Sales_Transaction;
SELECT * FROM Project;
SELECT * FROM Department_Manager_History;
SELECT * FROM Employee_Project;
SELECT * FROM Dependent;
SELECT * FROM Item_Transaction;
```

Figure 9 Tables (Item_Transaction, Dependent) INSERT Codes. COMMIT for Saving code changes. SELECT Statements for all tables.

SQL 1.2 Tables

BRANCHID	BRANCHNAME	CITY	ADDRESS
101	Granada Branch	Riyadh	Granada Mall, Gate 3
102	Corniche Branch	Dammam	Corniche Road
103	Nakheel Branch	Riyadh	Nakheel Mall, Gate 2

Table 1 Branch Table

DEPARTMENTID	DEPARTMENTNAME	NUMEMPLOYEES	BRANCHID
1	Store Operations	12	101
2	Cashiers	18	101
3	Customer Service	6	101
4	Store Operations	10	102
5	Cashiers	15	102
6	Customer Service	5	102
7	Store Operations	11	103
8	Cashiers	16	103
9	Marketing & Promotions	6	103

Table 2 Department Table

EID	FNAME	LNAME	POSITION	SALARY	ADDRESS	SEX	DOB	HIREDATE	DEPARTMENTID
201	Ahmed	Salem	Store Operations Manager	7500	Riyadh – Granada	M	12-APR-86	01-MAR-19	1
202	Rana	Yousif	Stock Clerk	4800	Riyadh – Granada	F	20-SEP-95	10-JUN-21	1
203	Majed	Saleh	Floor Staff	4600	Riyadh – Granada	M	05-NOV-97	15-AUG-22	1
204	Layla	Faris	Cashier Supervisor	6200	Riyadh – Granada	F	10-MAY-92	01-SEP-18	2
205	Saif	Hamad	Cashier	4500	Riyadh – Granada	M	22-JAN-98	14-FEB-21	2
206	Reem	Nasser	Cashier	4450	Riyadh – Granada	F	30-JUL-99	01-APR-22	2
207	Nora	Ali	CS Supervisor	6000	Riyadh – Granada	F	02-SEP-93	15-JAN-19	3
208	Huda	Ibrahim	CS Agent	4700	Riyadh – Granada	F	18-MAR-97	20-MAY-21	3
209	Sami	Kareem	CS Agent	4650	Riyadh – Granada	M	01-DEC-96	10-FEB-22	3
210	Faisal	Omar	Store Manager	7800	Dammam – Corniche	M	28-DEC-87	01-NOV-18	4
211	Ali	Zaid	Receiver	5000	Dammam – Corniche	M	09-JUN-94	05-JUL-20	4
212	Maha	Adel	Floor Staff	4700	Dammam – Corniche	F	09-FEB-98	12-MAR-22	4
213	Reema	Majed	Cashier Supervisor	6100	Khobar – Thuqbah	F	19-MAR-91	01-OCT-19	5
214	Tariq	Zayed	Cashier	4550	Dammam – Corniche	M	22-AUG-97	25-MAR-21	5
215	Lina	Saad	Cashier	4400	Dammam – Corniche	F	11-NOV-99	18-JUN-22	5
216	Abeer	Hassan	CS Supervisor	5900	Dammam – Corniche	F	05-OCT-90	10-AUG-18	6
217	Omar	Rashid	CS Agent	4700	Dammam – Corniche	M	14-MAY-96	30-SEP-20	6
218	Dana	Khalil	CS Agent	4650	Dammam – Corniche	F	27-SEP-97	22-NOV-21	6
219	Sami	Nasser	Operations Supervisor	7000	Riyadh – Nakheel	M	05-NOV-89	10-JAN-20	7
220	Hind	Mousa	Floor Staff	4800	Riyadh – Nakheel	F	07-APR-95	15-AUG-21	7
221	Yara	Samir	Inventory Clerk	4900	Riyadh – Nakheel	F	19-JAN-96	03-MAY-22	7
222	Nabil	Saleh	Cashier Supervisor	6150	Riyadh – Nakheel	M	16-FEB-91	20-JUN-19	8
223	Waleed	Anwar	Cashier	4550	Riyadh – Nakheel	M	09-SEP-98	27-JAN-21	8
224	Sara	Jalal	Cashier	4450	Riyadh – Nakheel	F	24-DEC-99	11-JUL-22	8
225	Aisha	Rahman	Marketing Coordinator	6400	Riyadh – Nakheel	F	12-SEP-88	20-FEB-19	9
226	Hani	Yusuf	Graphic Designer	5500	Riyadh – Nakheel	M	03-MAR-94	18-APR-21	9
227	Lujain	Hakim	Social Media Specialist	5300	Riyadh – Nakheel	F	29-AUG-95	05-JAN-22	9

Table 3 Employee Table

CUSTOMERID	FNAME	LNAME	PHONE	EMAIL
801	Khalid	Omar	0551122334	khalid.o@example.com
802	Mariam	Hassan	0569988776	mariam.h@example.com
803	Saad	Fahad	0547766554	saad.f@example.com

Table 4 Customer Table

CATEGORYID	CATEGORYNAME
1	Produce
2	Bakery
3	Dairy & Beverages
4	Household & Cleaning

Table 5 Category Table

PRODUCTID	PNAME	BRAND	PRICE	UNIT	CATEGORYID
701	Tomatoes	FreshFarms	4.5	Kg	1
702	White Bread Loaf	SunBake	7.5	Piece	2
703	Milk 2L	DairyPure	8.75	Liter	3
704	Floor Cleaner	CleanHome	12	Bottle	4

Table 6 Product Table

SUPPLIERID	SUPPLIERNAME	PHONE	CITY	BRANCHID
501	FreshFarms Co.	0501234567	Riyadh	101
502	Gulf Packaging Ltd.	0543342211	Dammam	102
503	CleanHome Products	0558899772	Riyadh	103

Table 7 Supplier Table

TRANSACTIONID	TRANSACTIONDATE	TOTALAMOUNT	EID	CUSTOMERID
1001	15-FEB-24	64.5	205	801
1002	17-FEB-24	22	214	802
1003	17-FEB-24	45.75	223	803

Table 8 Sales_Transaction Table

PROJECTID	PROJECTNAME	LOCATION	DEPARTMENTID
301	Aisle Reorganization	Granada Branch – Operations	1
302	Cashier Queue Optimization	Granada Branch – Checkout	2
303	Customer Feedback Kiosk Setup	Granada Branch – Entry Area	3
304	Inventory Layout Improvement	Corniche Branch – Warehouse	4
305	Express Cashier Lane Setup	Corniche Branch – Front End	5
306	Customer Inquiry System Upgrade	Corniche Branch – CS Desk	6
307	Storage Area Redesign	Nakheel Branch – Backroom	7
308	Cashier Scheduling Automation	Nakheel Branch – Cash Office	8
309	Weekend Deals Ad Campaign	Riyadh Region	9

Table 9 Project Table

STARTDATE	DEPARTMENTID	EID
01-MAR-19	1	201
01-SEP-18	2	204
15-JAN-19	3	207
01-NOV-18	4	210
01-OCT-19	5	213
10-AUG-18	6	216
10-JAN-20	7	219
20-JUN-19	8	222
20-FEB-19	9	225

Table 10 Department_Manager_History

ASSIGNMENTID	STARTDATE	ENDDATE	PROJECTID	EID
901	05-JAN-24	10-FEB-24	301	201
902	05-JAN-24	10-FEB-24	301	202
903	05-JAN-24	10-FEB-24	301	203
904	12-FEB-24	05-MAR-24	302	204
905	12-FEB-24	05-MAR-24	302	205
906	08-MAR-24	30-MAR-24	303	207
907	08-MAR-24	30-MAR-24	303	208
908	15-JAN-24	20-FEB-24	304	210
909	22-FEB-24	18-MAR-24	305	213
910	10-MAR-24	02-APR-24	306	216
911	20-JAN-24	25-FEB-24	307	219
912	27-FEB-24	22-MAR-24	308	222
913	01-FEB-24	28-FEB-24	309	225

Table 11 Employee_Project Table

DEPENDENTID	DEPENDENTNAME	SEX	RELATIONSHIP	DOB	EID
1	Sara Ahmed	F	Daughter	14-AUG-20	201
2	Mohammed Faisal	M	Son	01-MAR-15	210
3	Layan Aisha	F	Daughter	22-NOV-17	225

Table 12 Dependent Table

ITEMID	QUANTITY	TRANSACTIONID	PRODUCTID
1	2	1001	702
2	1	1001	703
3	3	1002	704
4	1	1003	701
5	2	1003	702

Table 13 Item_Transaction Table

SQL 1.3 Queries

```sql
SELECT e.fName, e.lName, d.departmentName
FROM Employee e
JOIN Department d ON e.departmentID = d.departmentID;
```

Query 1 Displays each employee along with the department they belong to.

FNAME	LNAME	DEPARTMENTNAME
Ahmed	Salem	Store Operations
Rana	Yousif	Store Operations
Majed	Saleh	Store Operations
Layla	Faris	Cashiers
Saif	Hamad	Cashiers
Reem	Nasser	Cashiers
Nora	Ali	Customer Service
Huda	Ibrahim	Customer Service
Sami	Kareem	Customer Service
Faisal	Omar	Store Operations
Ali	Zaid	Store Operations
Maha	Adel	Store Operations
Reema	Majed	Cashiers
Tariq	Zayed	Cashiers
Lina	Saad	Cashiers
Abeer	Hassan	Customer Service
Omar	Rashid	Customer Service
Dana	Khalil	Customer Service
Sami	Nasser	Store Operations
Hind	Mousa	Store Operations
Yara	Samir	Store Operations
Nabil	Saleh	Cashiers
Waleed	Anwar	Cashiers
Sara	Jalal	Cashiers
Aisha	Rahman	Marketing & Promotions
Hani	Yusuf	Marketing & Promotions
Lujain	Hakim	Marketing & Promotions

Query Result 1

```sql
SELECT d.departmentName, COUNT(e.eID) AS numEmployees
FROM Department d
LEFT JOIN Employee e ON d.departmentID = e.departmentID
GROUP BY d.departmentName;
```

Query 2 Shows the total number of employees in each department.

DEPARTMENTNAME	NUMEMPLOYEES
Marketing & Promotions	3
Cashiers	9
Customer Service	6
Store Operations	9

Query Result 2

```
SELECT p.projectName, d.departmentName
FROM Project p
JOIN Department d ON p.departmentID = d.departmentID;
```

Query 3 Displays projects and the departments responsible for them.

PROJECTNAME	DEPARTMENTNAME
Aisle Reorganization	Store Operations
Cashier Queue Optimization	Cashiers
Customer Feedback Kiosk Setup	Customer Service
Inventory Layout Improvement	Store Operations
Express Cashier Lane Setup	Cashiers
Customer Inquiry System Upgrade	Customer Service
Storage Area Redesign	Store Operations
Cashier Scheduling Automation	Cashiers
Weekend Deals Ad Campaign	Marketing & Promotions

Query Result 3

```
SELECT e.fName, e.lName, p.projectName
FROM Employee_Project ep
JOIN Employee e ON ep.eID = e.eID
JOIN Project p ON ep.projectID = p.projectID;
```

Query 4 Lists employees and the projects they are assigned to.

FNAME	LNAME	PROJECTNAME
Ahmed	Salem	Aisle Reorganization
Rana	Yousif	Aisle Reorganization
Majed	Saleh	Aisle Reorganization
Layla	Faris	Cashier Queue Optimization
Saif	Hamad	Cashier Queue Optimization
Nora	Ali	Customer Feedback Kiosk Setup
Huda	Ibrahim	Customer Feedback Kiosk Setup
Faisal	Omar	Inventory Layout Improvement
Reema	Majed	Express Cashier Lane Setup
Abeer	Hassan	Customer Inquiry System Upgrade
Sami	Nasser	Storage Area Redesign
Nabil	Saleh	Cashier Scheduling Automation
Aisha	Rahman	Weekend Deals Ad Campaign

Query Result 4

```
19  SELECT c.fName, c.lName, t.transactionID, t.totalAmount
20  FROM Customer c
21  JOIN Sales_Transaction t ON c.customerID = t.customerID;
```

Query 5 Displays customer transactions and their total amounts.

FNAME	LNAME	TRANSACTIONID	TOTALAMOUNT
Khalid	Omar	1001	64.5
Mariam	Hassan	1002	22
Saad	Fahad	1003	45.75

Query Result 5

```
25  SELECT t.transactionID, p.pName, i.quantity
26  FROM Sales_Transaction t
27  JOIN Item_Transaction i ON t.transactionID = i.transactionID
28  JOIN Product p ON i.productID = p.productID;
```

Query 6 Shows which products were sold in each transaction.

TRANSACTIONID	PNAME	QUANTITY
1001	White Bread Loaf	2
1001	Milk 2L	1
1002	Floor Cleaner	3
1003	Tomatoes	1
1003	White Bread Loaf	2

Query Result 6

```
32 v SELECT e.fName, e.lName, SUM(t.totalAmount) AS totalSales
33   FROM Employee e
34   JOIN Sales_Transaction t ON e.eID = t.eID
35   GROUP BY e.fName, e.lName;
```

Query 7 Calculates the total sales amount handled by each employee.

FNAME	LNAME	TOTALSALES
Waleed	Anwar	45.75
Tariq	Zayed	22
Saif	Hamad	64.5

Query Result 7

```
39 v SELECT s.supplierName, b.branchName
40   FROM Supplier s
41   JOIN Branch b ON s.branchID = b.branchID;
```

Query 8 Displays suppliers and the branches they supply.

SUPPLIERNAME	BRANCHNAME
FreshFarms Co.	Granada Branch
Gulf Packaging Ltd.	Corniche Branch
CleanHome Products	Nakheel Branch

Query Result 8

```
45 v SELECT e.fName, e.lName, d.dependentName
46   FROM Employee e
47   JOIN Dependent d ON e.eID = d.eID;
```

Query 9 Lists employees along with their registered dependents.

FNAME	LNAME	DEPENDENTNAME
Ahmed	Salem	Sara Ahmed
Faisal	Omar	Mohammed Faisal
Aisha	Rahman	Layan Aisha

Query Result 9

```
51 v SELECT d.departmentName, e.fName, e.lName, m.startDate
52   FROM Department_Manager_History m
53   JOIN Department d ON m.departmentID = d.departmentID
54   JOIN Employee e ON m.eID = e.eID;
```

Query 10 Shows department managers and the dates they were appointed.

DEPARTMENTNAME	FNAME	LNAME	STARTDATE
Store Operations	Ahmed	Salem	01-MAR-19
Cashiers	Layla	Faris	01-SEP-18
Customer Service	Nora	Ali	15-JAN-19
Store Operations	Faisal	Omar	01-NOV-18
Cashiers	Reema	Majed	01-OCT-19
Customer Service	Abeer	Hassan	10-AUG-18
Store Operations	Sami	Nasser	10-JAN-20
Cashiers	Nabil	Saleh	20-JUN-19
Marketing & Promotions	Aisha	Rahman	20-FEB-19

Query Result 10

These queries demonstrate how the database can efficiently retrieve and analyze information across multiple related tables. They confirm the correctness of relationships and the effectiveness of the database design in supporting real business operations.

Conclusion

This project has shown the successful development of a relational database system for AlTamimi Supermarkets. The solution did not only simplify the complex process related to the large retail organization's branches, departments, employees, suppliers, products, customers, and sales transactions but indulged them through the planting of the right requirements analysis. An Entity Relationship Diagram (ERD) was drawn that mirrored the

business's structure and allowed one-to-many and many-to-many relations, embracing employee project assignments and sales transactions.

Normalization of the database schema was conducted methodically and up to the Third Normal Form (3NF), the process that maintained data integrity, eliminated duplicates, and avoided anomalies of update, insertion, and deletion. Logical separations of key entities like Employee, Department, Sales_Transaction, Product, were done; at the same time, primary and foreign keys were imposed to enforce strong referential integrity all across the system. The database was so much enhanced by the addition of historical tables like Department_Manager_History that it became possible for the organization to keep track of the changes in the management over a period.

In general, the suggested database design is up to all the project requirements and offers a structured, scalable, and effective data management solution that facilitates tracking of operations, making decisions and expanding the system in the future. The project, moreover, is an illustration of the practical use of database management concepts, like ER modeling, normalization, and SQL-based implementation, in a real business setting.

Though the current database design is effective for Al-Tamimi Supermarkets operationally, it still leaves room for improvements in the future:

1.Advanced Security Controls

To secure the sensitive employee and financial data, implement role-based access control that only allows managers, cashiers, and administrators to see it as per their user roles.

2.Reporting and Analytics Integration

By integrating business intelligence tools, the production of various reports such as sales, employee’s performance, and branch-level analysis will be made easier thus aiding in making better strategic decisions.

3.Inventory Management Extension

Real-time monitoring of inventory tracking tables could be implemented to track stock levels, reorder points, and supplier delivery schedules.

4.System Automation and Triggers

Database triggers and stored procedures could be used for updating the number of employees categorized by department, as well as total amounts of transactions, among other things, automatically.

5.Scalability and Cloud Deployment

To manage scalability, high availability, and improved performance as the business continues to grow, the database could be implemented on a cloud-based DBMS.

Through the adoption of these recommendations, Al-Tamimi Supermarkets will not only increase the effectiveness of its database system but also its security and analytical capabilities, thus securing operational efficiency and competitive advantage in the long run.